
SYSTEM OVERVIEW

System Type:

Circumbinary Star System

Two stars orbit one another.

All planets orbit the binary pair.

Purpose:

Provide a stable, believable astrophysical
foundation for MK147 while preserving:

- Two Suns
- Twin Sunrises
- Twin Sunsets
- Long Days
- Stable Agriculture
- Stable Civilization

THE STARS

Star A

Mass:
0.55 Solar Mass

Star B

Mass:
0.45 Solar Mass

Combined System

≈ 1 Solar Mass Equivalent

Both stars orbit a shared barycenter.

The stars remain close enough that the
planetary system experiences them almost
as a single gravitational body.

BINARY ORBIT

Star Separation:

≈ 0.25 AU

Binary Orbital Period:

≈ 46 Days

Meaning:

Every 46 planetary days the two stars
complete one mutual orbit.

This creates:

- Variable sky positions
- Twin-sun phenomena
- Seasonal visual variation

without destabilizing the planets.

PLANET I

Designation:

The Ash World

Orbital Radius:

0.60 AU

Orbital Period:

≈ 170 Days

Environment:

Extremely Hot

Mercury/Venus Hybrid

Minimal Civilization

Primarily Background Lore

PLANET II

Designation:

Kira's World

Orbital Radius:

1.20 AU

Orbital Period:

≈ 360 Days

Primary Setting

Goldilocks World

Environment:

- Oceans
- Forests
- Agriculture
- Kingdoms
- Universities
- Blackstone

All major story events occur here.

PLANET III

Designation:

The Frontier World

Orbital Radius:

2.40 AU

Orbital Period:

≈ 1,020 Days

Environment:

Cold

Sparse

Difficult

Research Colonies

Exploration Focused

PLANET IV

Designation:

The Great Giant

Orbital Radius:

5.50 AU

Orbital Period:

≈ 4,720 Days

Environment:

Gas Giant

Possible Rings

Visible From Kira's World

Major Astronomical Object

KIRA'S WORLD
PLANETARY ROTATION

Length of Day:

31 Hours

Canonical

Purpose:

Narrative Organization

Travel Organization

Educational Organization

Civilizational Organization

KIRA'S YEAR

Orbital Period:

360 Days

Each Day:

31 Hours

Total Year Length:

11,160 Hours

To inhabitants:

Simply:

One Year

No one thinks in Earth time.

SEASONS

Four Seasons

Spring

Summer

Autumn

Winter

Each Season:

90 Days

Seasonal Structure:

Spring

=

90 Days

Summer

=

90 Days

Autumn

=

90 Days

Winter

=

90 Days

BLACKSTONE CALENDAR

Semester I

Spring

90 Days

Semester II

Summer

90 Days

Semester III

Autumn

90 Days

Winter

Break

90 Days

CELESTIAL PHENOMENA

Common:

- Twin Sunrises
- Twin Sunsets
- Dual Shadows
- Extended Twilight

Periodic:

Every 46 Days

Binary Alignment Shift

Observable by astronomers.

Rare:

Every Few Years

Major Celestial Events

Examples:

- Double Eclipses
- Twin-Star Conjunctions
- Giant Planet Opposition
- Ring Visibility Events

ASTRONOMICAL HIERARCHY

Binary Orbit

46 Days

↓

Planetary Day

31 Hours

↓

Season

90 Days

↓

Year

360 Days

↓

Civilization

↓

Kingdom

↓

University

↓

Student

↓

Story

AUTHOR NOTE

This system is intended to be:

Stable

Believable

Internally Consistent

Easy To Write

The purpose is not to maximize
astronomical complexity.

The purpose is to create a sky
that naturally supports:

Kira

Blackstone University

The Kingdom

The Knowledge Era

And all future MK147 stories.